SHASHWAT DWIVEDI

Backend Software Engineer – Distributed Systems & AI/LLM Applications

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Summary

Backend Software Engineer with 2+ years of experience building scalable, distributed backend systems and AI/LLM-powered applications. Skilled in Node.js, PostgreSQL, Kafka, Redis, and Retrieval-Augmented Generation (RAG), with a track record of improving latency, throughput, and reliability in production systems serving thousands of endpoints and users.

Professional Experience

Techbridge Consultancy Services Pvt. Ltd. **May 2024 – Present**
Software Engineer *Gurugram, India*

- Own backend architecture for enterprise security products spanning Patch Management and Vulnerability Assessment, building distributed scan orchestration and real-time telemetry across **thousands of endpoints** using Node.js, PostgreSQL, and Server-Sent Events (SSE).
- Engineered a high-throughput vulnerability intelligence ingestion pipeline using asynchronous processing and batched writes, reducing **detection latency by 40%** under production load.
- Reduced **p95 API latency by 55%** through PostgreSQL query restructuring, composite indexing, and connection-pool optimization.
- Introduced **Kafka-based event-driven microservices** to decouple scan orchestration into independently scalable services and improve fault isolation.
- Built a Windows desktop agent that collects OS, hardware, software, and security metadata from endpoints and streams telemetry to backend REST APIs, replacing manual collection workflows.
- Led **SSO integration across 4+ internal applications** using OAuth 2.0 and SAML, centralizing authentication and role-based access control (RBAC).

Abekus EdTech Platform **Feb 2023 – Sep 2023**
Software Engineering Intern *Noida, India*

- Developed backend services using Node.js and PostgreSQL and introduced Redis caching on high-traffic endpoints, reducing average API response time by **30%** under concurrent load.
- Eliminated N+1 queries and added missing database indexes across critical user flows, reducing query execution time by approximately **35%** for a platform serving **50,000+ active users**.
- Partnered with product and engineering teams to resolve scalability bottlenecks ahead of peak exam-season traffic, achieving **zero downtime at peak**.

Projects

Memora — Enterprise AI Knowledge Assistant GitHub
Python, FastAPI, Node.js, NestJS, PostgreSQL, pgvector, Redis, BullMQ, OpenAI API, RAG, BM25, SSE

- Built a multi-tenant enterprise AI knowledge assistant with document ingestion, embedding generation, hybrid retrieval, and context-aware responses using Retrieval-Augmented Generation (RAG).
- Designed an extensible connector architecture for Google Drive, GitHub, Notion, Confluence, Jira, Slack, SharePoint, and PostgreSQL with incremental synchronization.

Cross-Platform Vulnerability Scanner GitHub
C++, Go, Nmap, Socket Programming, Multithreading

- Built a multithreaded network port scanner using raw socket programming, achieving **500+ ports/sec** with configurable scan depth and rate limiting.
- Added in-memory caching for CVE and vulnerability intelligence lookups, reducing redundant database queries by **60%** during large scans.

Technical Skills

Languages: JavaScript, Python, C++
Backend & APIs: Node.js, Express.js, FastAPI, NestJS, REST APIs, GraphQL, WebSockets, SSE
Distributed Systems: Apache Kafka, Redis, BullMQ, Event-Driven Architecture, Microservices
Databases: PostgreSQL, pgvector
AI/LLM: OpenAI API, LangChain, RAG, Vector Embeddings, BM25, Llama 3
Cloud & DevOps: AWS (EC2, S3), Docker, CI/CD, Linux, Git

Achievements

- **1st Place, TechBridge Innovathon 2025:** Led a team building an AI-powered SIEM chatbot with LLM integration and real-time log analysis; selected best project among **30+ teams**.
- **1st Runner-Up, Rajasthan IT Day Hackathon 2023:** Built a government-facing solution at a state-level hackathon against **500+ competing teams**.
- **Codeforces Specialist:** 1400+ max rating; solved **450+ DSA problems** on Codeforces and LeetCode.

Education

Ajay Kumar Garg Engineering College **2020 – 2024**
B.Tech., Computer Science & Engineering *Ghaziabad, India*